

Technische Hochschule Ingolstadt
Fakultät Elektro- und Informationstechnik

Sensor Network Technologies and Sensor Fusion

Variance and Covariance
Covariance as Linear Transform

Nov 5, 2025

Welcome to the lecture!

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Technische Hochschule Ingolstadt

AIMotion, AI and Sensor Data Fusion

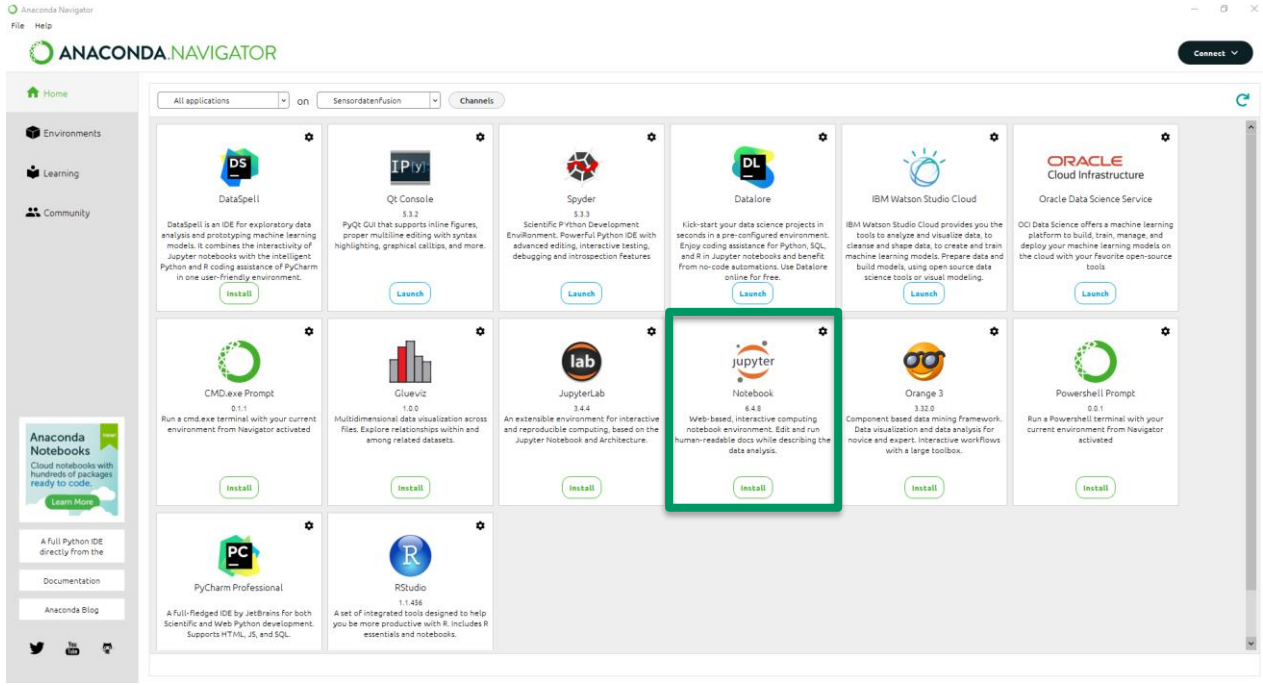
Email: bjoern.giesler@thi.de

- **Please register for the lecture on Moodle!**
  - Link: https://moodle.thi.de/course/view.php?id=10580
  - Key: **SensNT+DF#WS25**
- **News on Moodle:**
  - Slides for the lecture (upcoming lectures marked as WIP and are subject to change!)
  - Formulary, Exercises (Solutions provided after discussion in the lecture)
  - Exercises are optional, one task / sheet per week, but are highly recommended.
    They can be handed in for correction.
  - I will *try* to publish exercises before Wednesday‘s lecture so that initial questions can be discussed on Thursday.
- **Exam: Written exam, 90 minutes**
- **Consultation hours: Please arrange via e-mail to bjoern.giesler@thi.de.**
- **Please also don‘t hesitate to send questions, lecture feedback, change requests, etc.!**

- **You will find two exercises accompanying last week‘s lectures on Moodle.**
- ***Exercise Sheet 1***
  - A couple of very gentle initial exercises (vector/matrix multiplication, Euler angles, derivatives / gradient vectors).
  - Doing this sheet is **highly recommended**, as is handing it in for correction. Depending on number of submissions, I will try to give results on Monday.
- ***ProgExercise 2***
  - A Python program which realizes a Viola Jones face detector.
  - Your task is to use the supplied camera calibration matrix K, and the size of the detected face, to estimate the 3D position of the face in space.
  - This will develop into a longer series of programming exercises estimating positions and velocities of objects in space with Kalman Filters.
  - Doing the exercise therefore is **highly recommended** as well.
- **Solutions for both of the exercises will be posted on Wed Oct 22.**

Software: Python/Anaconda

- For the ML examples we will use today, will use the Jupyter Notebook environment.
- It is recommended you install the Anaconda Python environment and distribution, that installs Jupyter as one of its environments.
- You can find it at https://www.anaconda.com/.



Let‘s look at the...

- **...exercise Sheet 2: Mean, Variance, Standard Deviation, Eigenvalues, Covariance Matrices**
- **...coupled programming exercise!**

We are trying something new today – **Voting Cups!**

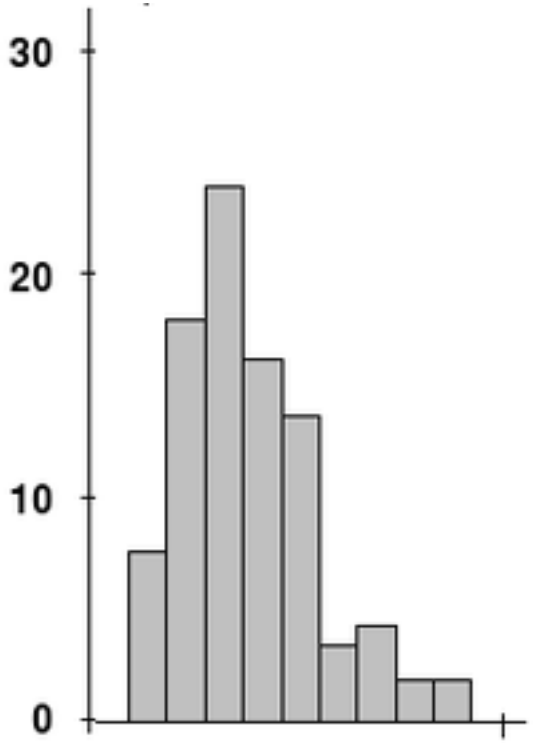
- This will help me understand and tailor explanation speed to your needs.
- Each lecture, if you want to participate, grab a set of 4 cups from the lecturer‘s desk.
- Each set has 4 colors. Stack the cups to the color that reflects your current ability to follow.



Moments and Central Moments of Probability Distributions

- **Moments of a function: Quantitative measures related to the shape of the function's graph.**
  - If the function is a probability distribution, then
    - the first moment is the expected value,
    - the second central moment is the variance,
    - the third standardized moment is the skewness (0 for symmetric distributions).

Moment ordinal	Moment		
	Raw	Central	Standardized
1	Mean	0	0
2	—	Variance	1
3	—	—	Skewness



- **Moments are calculated**
  - **raw:** around zero (i.e. without regard for the mean of the distribution)
  - **central:** around the mean
  - **standardized:** around the mean and divided by the variance

Moments of a Probability Distribution

- The i-th **moment** (non-central, non-standardized) of a probability distribution is computed as

$$E\left[X^i\right] = \int_{-\infty}^{\infty} x^i \cdot p_X(x)dx \quad \text{or} \quad E\left[X^i\right] = \sum_{k\in\mathcal{N}} x(k)^i \cdot p_X(x(k))$$

- Example for our dice throw: $E[\text{dice}] = \frac{1}{6} + \frac{2}{6} + \frac{3}{6} + \frac{4}{6} + \frac{5}{6} + \frac{6}{6} = \frac{21}{6} = 3.5$

- The first moment is denoted as mean $\boldsymbol{\mu}$.

$$\mu \equiv E[X]$$

Moments of a Probability Distribution

- **Central moments** are calculated by subtracting the mean first.

  The i-th **central moment** of a probability distribution is

$$\begin{aligned} & E[(X - E[X])^i] \\ = & E[(X - \mu)^i] \\ = & \int_{-\infty}^{\infty} (x - \mu)^i \cdot p_X(x) dx \end{aligned}$$

  or $\sum_{k \in N} (x(k) - \mu)^i \cdot p_X(x(k))$ for the discrete case.

- The **second central moment is called variance** $\mathrm{Var}(X)$**,** and it square root is called the **standard deviation.**

$$\mathrm{Var}[X] = \sigma^2 = E[(X - E[X])]^2 = E[X^2] - E[X]^2$$ and $$\sigma = \sqrt{\mathrm{Var}[X]}$$

Computational Rules for the Mean

- For two random variables X and Y as well as constants a and b the following rules hold:

$$\begin{aligned} E[a] &= a, \\ E[aX] &= aE[X], \\ E[a+bX] &= a + bE[X], \\ E[X+Y] &= E[X] + E[Y] \end{aligned}$$

and for stochastically independent random variables X and Y

$$E[X \cdot Y] = E[X] \cdot E[Y]$$

holds.

Computational Rules for the Variance

- For two random variables X and Y as well as constants a and b the following rules hold:

$$\begin{aligned} \mathrm{Var}[a] &= 0, \\ \mathrm{Var}[a+X] &= \mathrm{Var}[X], \\ \mathrm{Var}[a+bX] &= b^2 \cdot \mathrm{Var}[X], \\ \mathrm{Var}[X+Y] &= \mathrm{Var}[X] + \mathrm{Var}[Y] \\ &\quad + 2 \cdot \underbrace{(E[X \cdot Y] - E[X]\,E[Y])}_{\mathrm{Cov}(X,Y)} \end{aligned}$$

and for statistically independent random variables X and Y

$$\mathrm{Var}[X+Y] = \mathrm{Var}[X] + \mathrm{Var}[Y]$$

holds.

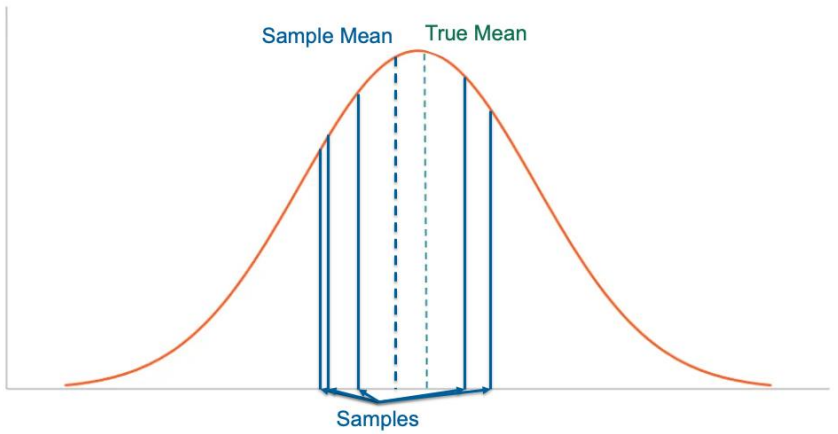
Population and Sample Mean and Variance

- **Our rules so far spoke of *probability density functions* which (in the discrete case) determine the likelihood of each event x_i in the population X.**
- **When we try to determine the mean and variance for the PDF, we**
  - sample N values $x_1, \ldots, x_n$ randomly
  - calculate the mean $\bar{x}$ and variance s^2 as estimates for μ, σ^2 .
  - Knowing nothing about the distribution, we have to assume $p_X(X_i) = 1/N$ for each sample.
- **With $\sigma^2 = E[(X-\mu)^2] = \sum (x_i - \mu)^2 p_X(x_i)$, we could assume $s^2 = \frac{1}{N}\sum (x_i - \bar{x})^2$ for the variance.**
- **However this would result in an estimate biased towards the mean.**
- **The unbiased term is obtained by multiplying with *Bessel's Correction* $N/(N-1)$*:***

$$s^2 = \frac{1}{N-1}\sum (x_i - \bar{x})^2.$$

Bessel‘s Correction

- **Intuitive explanation:**
  - By pulling a sample set, we don‘t calculate the actual population mean μ but the mean of the sample set $\bar{x}$.
  - Therefore on average $\sum(x_i - \bar{x})^2$ can be expected to be smaller than $\sum(x_i - \mu)^2$ - we will be consistently underestimating the variance.
  - Dividing by $N - 1$ instead of N introduces a small correction factor.
- **For large sample sets or the whole population ($N \to \infty$), $N/(N-1)$ converges to 1 and we have the original term.**
- **The true mathematical derivation of Bessel‘s Correction is quite involved.**
- **There are a number of examples on the Internet for self-study, e.g. [1]**

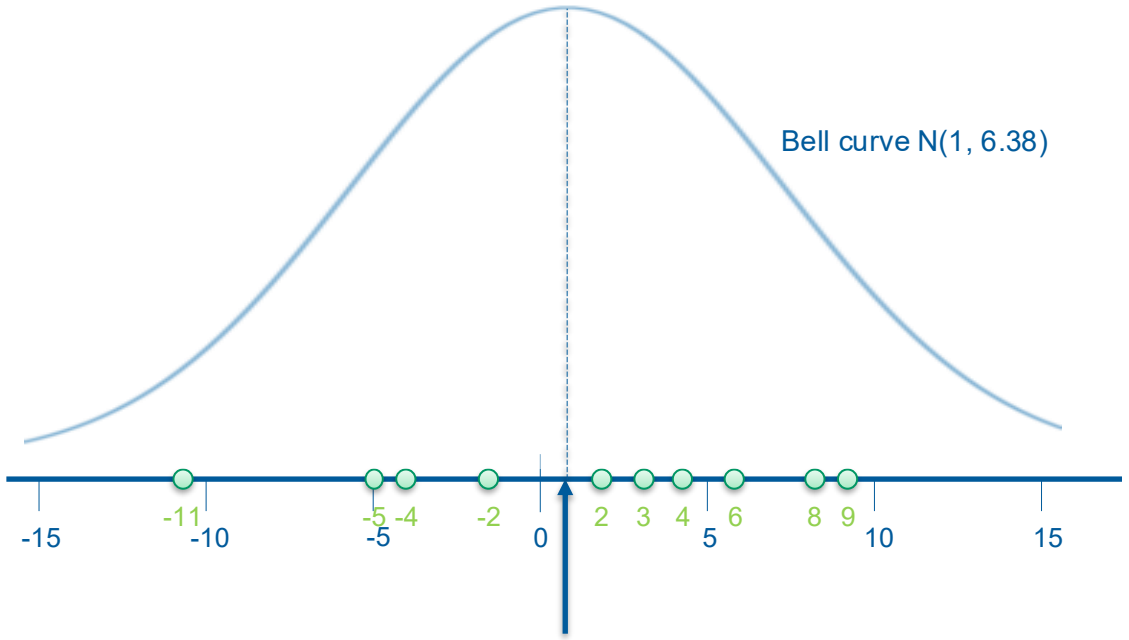


Sensor Networks and Sensor Fusion

(Central) Moments of Probability Distributions

Example:

- We record a set of 10 data points:
  X={3, -5, 4, 8, -11, -4, -2, 9, 6, 2}
- Since we have no a priori knowledge,
  we assume $P(X) = 1/10$ for all points.



Expectation: $$\bar{x} = \sum_i x_i \cdot p_X(x_i) = \frac{3 - 5 + 4 + 8 - 11 - 4 - 2 + 9 + 6 + 2}{10} = \frac{10}{10} = 1$$

Variance: $$s_X^2 = \frac{1}{N-1} \sum_i (x_i - \bar{x})^2 = \frac{2^2 + (-6)^2 + 3^2 + 7^2 + (-12)^2 + (-5)^2 + (-3)^2 + 8^2 + 5^2 + 1^2}{9} = \frac{366}{9} \approx 40.67$$

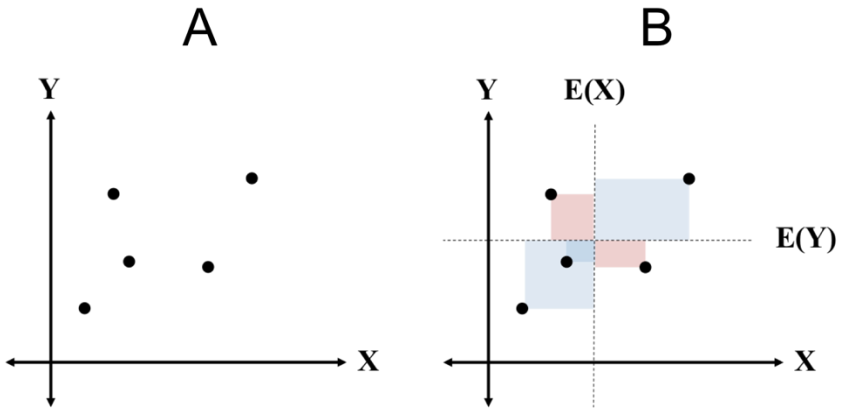
Standard Deviation: $$s = \sqrt{40.67} \approx 6.38$$

Co-Variance

Covariance $Cov(X, Y)$

- Is a measure for the stochastic dependency between X and Y defined as

$$\begin{aligned}\mathrm{Cov}(X,Y) &= E\left[(X - E[X])\,(Y - E[Y])\right] \\ &= E[X \cdot Y] - 2 \cdot E[X] \cdot E[Y] + E[X] \cdot E[Y] \\ &= E[X \cdot Y] - E[X] \cdot E[Y].\end{aligned}$$



Visualization of the covariance

- Computational rules for the covariance

$$\mathrm{Cov}(X, X) = \mathrm{Var}(X),$$
$$\mathrm{Cov}(a + bX, c + dY) = b \cdot d \cdot \mathrm{Cov}(X, Y),$$
$$\mathrm{Cov}(X + Y, Z) = \mathrm{Cov}(X, Z) + \mathrm{Cov}(Y, Z)$$

- The **correlation coefficient** (normalized with values between -1 and $+1$) measures the degree of correlation between the two variables:

$$r_{XY} = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}[X] \cdot \mathrm{Var}[Y]}}$$

so a high stochastic dependency means $|r_{XY}|$ being close to 1. Independence means $r_{XY} \approx 0$.

Example Covariance - 1

- Two Sensors measure 11 values, contained in the following table

	1	2	3	4	5	6	7	8	9	10	11
Sensor $1/X$	2.6	1.0	3.6	0.5	1.6	1.9	2.9	2.2	0.9	2.8	2.0
Sensor $2/Y$	5	2	7	1	4	3	5	3	3	6	5

- Approximate the mean and variance as follows
  - Mean

$$E[X] = \overline{x} = \frac{1}{N}\sum_{k=1}^{N} x(k)$$

  - and variance

$$\mathrm{Var}[X] = \frac{1}{N-1}\sum_{k=1}^{N}(x(k) - E[X])^2$$

and compute the covariance $Cov(X, Y)$

Example Covariance - 2

	1	2	3	4	5	6	7	8	9	10	11	
Sensor $1/X$	2.6	1.0	3.6	0.5	1.6	1.9	2.9	2.2	0.9	2.8	2.0	Sum: 22
Sensor $2/Y$	5	2	7	1	4	3	5	3	3	6	5	Sum: 44

- We obtain the mean

$$E[X] = \frac{1}{11}\sum_{k=1}^{11} x(k) = 2, \qquad E[Y] = \frac{1}{11}\sum_{k=1}^{11} y(k) = 4$$

- and the variances

$$\begin{aligned}\mathrm{Var}[X] &= \frac{1}{10}\sum_{k=1}^{N}(x(k) - E[X])^2 \\ &= \frac{0.6^2 + 1^2 + 1.6^2 + 1.5^2 + 0.4^2 + 0.1^2 + 0.9^2 + 0.2^2 + 1.1^2 + 0.8^2 + 0^2}{10} = 0.904,\end{aligned}$$

$$\begin{aligned}\mathrm{Var}[Y] &= \frac{1}{10}\sum_{k=1}^{N}(y(k) - E[Y])^2 \\ &= \frac{1^2 + 2^2 + 3^2 + 3^2 + 0^2 + 1^2 + 1^2 + 1^2 + 1^2 + 2^2 + 1^2}{10} = 3.2.\end{aligned}$$

Example Covariance - 3

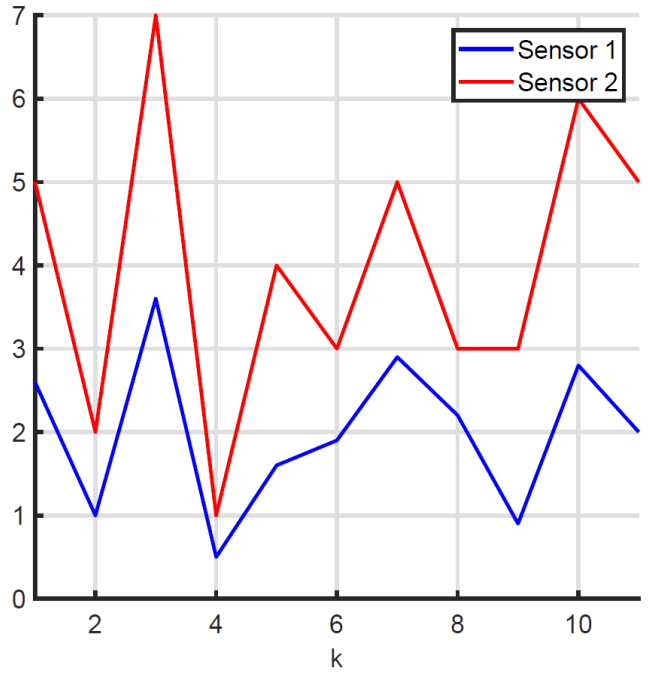
- Covariance

$$\mathrm{Cov}\,(X,Y) = \frac{1}{N-1}\sum_{k=1}^{N}\left(x(k) - E\,[X]\right) \cdot \left(y(k) - E\,[Y]\right)$$

$$= \frac{1}{10}\sum_{k=1}^{N}\left(x(k) - E\,[X]\right) \cdot \left(y(k) - E\,[Y]\right)$$

- The correlation coefficient is

$$r_{XY} = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}[X]\cdot\mathrm{Var}[Y]}} = \frac{1.54}{\sqrt{0.904\cdot 3.2}} = 0.9$$



Plot of the sensor measurements

Covariance

- **For generalized multivariate random variables, the expectation (= mean) is**

$$E[\mathbf{x}] = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} \vec{x} p(\mathbf{x} = \vec{x}) dx_1 \dots dx_n$$

or for the discrete case

$$E[\mathbf{x}] = \sum \sum \cdots \sum_{n} \vec{x} \cdot P(x_1, x_2, \dots, x_n)$$

- **The Covariance, which**
  - for two variables is $\mathrm{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$,
  - becomes the **Covariance Matrix** $\mathrm{Cov}_{\mathbf{xx}} = E[(\mathbf{x} - E(\mathbf{x}))(\mathbf{x} - E(\mathbf{x}))^{\top}]$.

Covariance

$$\begin{bmatrix} x_1 - \mu_1 \\ x_2 - \mu_2 \\ \vdots \\ x_n - \mu_n \end{bmatrix} \begin{bmatrix} x_1 - \mu_1 & x_2 - \mu_2 & \dots & x_n - \mu_n \end{bmatrix}$$

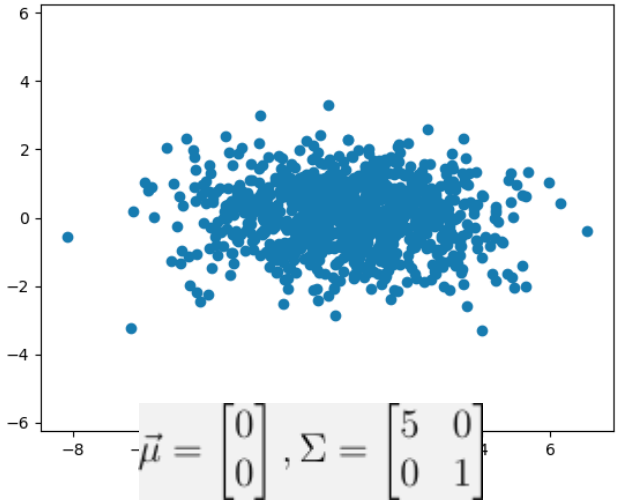
$$= \begin{bmatrix} (x_1 - \mu_1)^2 & (x_1 - \mu_1)(x_2 - \mu_2) & \dots & (x_1 - \mu_1)(x_n - \mu_n) \\ (x_2 - \mu_2)(x_1 - \mu_1) & (x_2 - \mu_2)^2 & \dots & (x_2 - \mu_2)(x_n - \mu_n) \\ \vdots & \vdots & \ddots & \vdots \\ (x_n - \mu_n)(x_1 - \mu_1) & (x_n - \mu_n)(x_2 - \mu_2) & \dots & (x_n - \mu_n)^2 \end{bmatrix}$$

$$= \begin{bmatrix} \sigma^2_{x_1} & \sigma^2_{x_1,x_2} & \dots & \sigma^2_{x_1,x_n} \\ \sigma^2_{x_2,x_1} & \sigma^2_{x_2} & \dots & \sigma^2_{x_2,x_n} \\ \dots & \dots & \ddots & \vdots \\ \sigma^2_{x_n,x_1} & \sigma^2_{x_n,x_2} & \dots & \sigma^2_{x_n} \end{bmatrix}$$

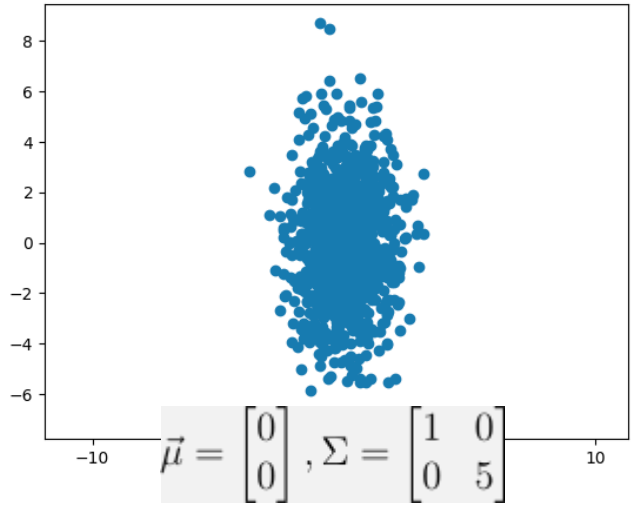
Some properties of the Covariance matrix...

- Always square.
- Diagonal elements are the variances of the vector elements.
- Elements to the side of the diagonal are the co-variances.
- Symmetric (i.e. $\mathrm{Cov}^\top = \mathrm{Cov}$).
- Positive semi-definite (i.e. $\mathbf{x}\mathrm{Cov}\mathbf{x}^\top \geq 0 \forall \mathbf{x} \neq \mathbf{0}$ -- the real-valued product of any nonzero vector with the matrix and the vector‘s transpose is larger or equal zero).
- Positive definite and invertible if all x_i have variance != 0.

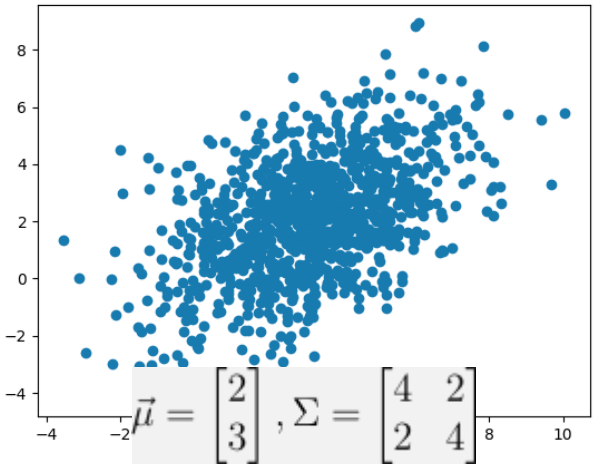
Scatter Plots of Normal Distributions with different means and covariances



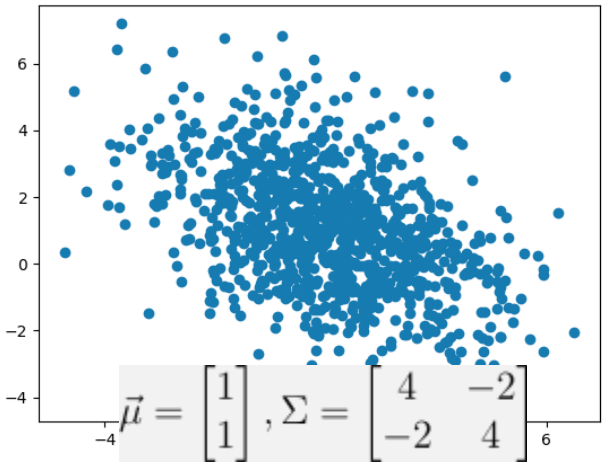
$$\vec{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma = \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}$$



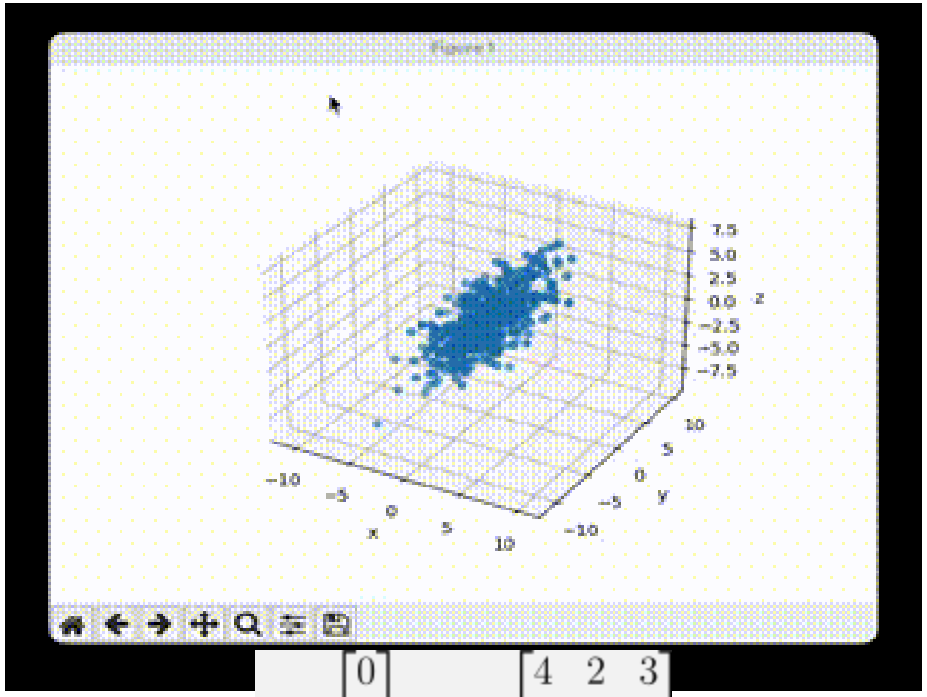
$$\vec{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$$



$$\vec{\mu} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$$



$$\vec{\mu} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \Sigma = \begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix}$$



$$\vec{\mu} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \Sigma = \begin{bmatrix} 4 & 2 & 3 \\ 2 & 5 & 3 \\ 3 & 3 & 6 \end{bmatrix}$$

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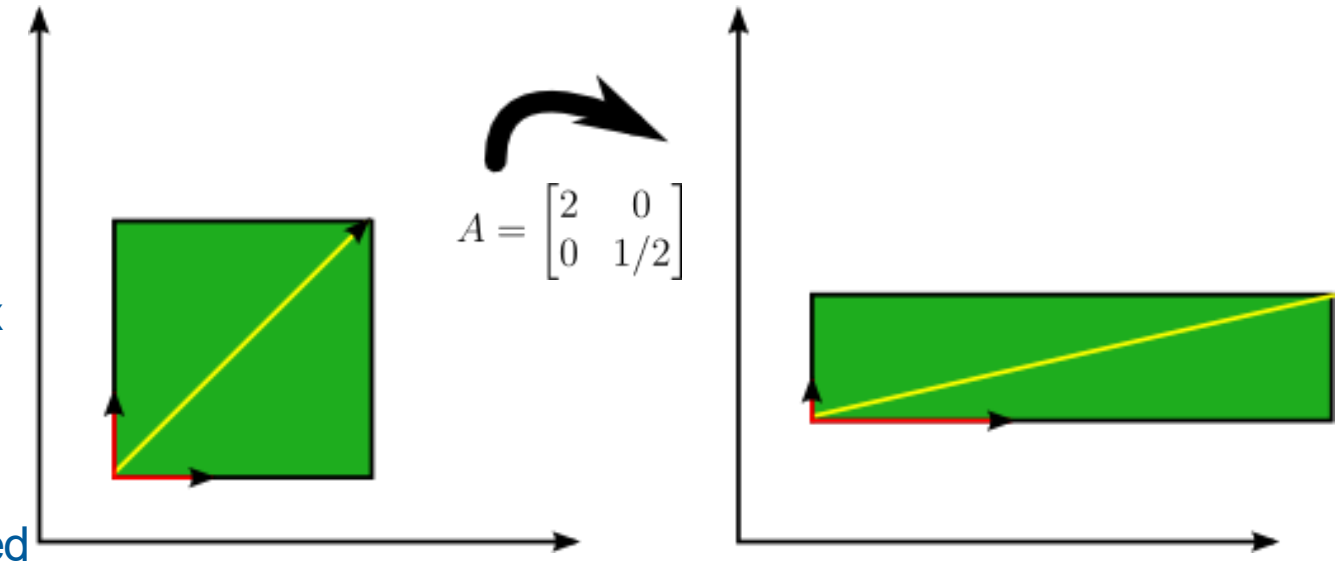
Co-Variance and Eigenvalues

Revisiting Eigenvectors and Eigenvalues

- *Eigenvectors* / *eigenvalues* have many important applications in machine learning and computer vision.
- An eigenvector is a vector whose direction remains unchanged when a linear transformation is applied.
- Consider the scaling transform $\vec{v}' = A\vec{v}$ on the right. The yellow vector changes direction, the red vectors do not.
- Therefore the red vectors are called eigenvectors of A.
- Generally, the eigenvectors $\vec{v}_i$ of a matrix A are vectors that satisfy

$$A\vec{v}_i = \lambda_i \vec{v}_i$$

where λ_i are corresponding scalars called eigenvalues.



Revisiting Eigenvectors and Eigenvalues

- We can rewrite the eigenvector equation from the next slide as follows:

$$A\vec{v}_i = \lambda_i \vec{v}_i$$
$$A\vec{v}_i - \lambda_i \vec{v}_i = 0$$
$$\vec{v}_i(A - \lambda_i I) = 0$$

  where I is the identity matrix of the same dimensions as A.

- This is of course true for the trivial solution of $\vec{v}_i = \vec{0}$.
- If $\vec{v}_i$ is not the null vector, $\vec{v}_i(A - \lambda_i I) = 0$ can only be defined if $(A - \lambda_i I)$ is not invertible.
- For a square matrix not to be invertible, its determinant must be zero.
- Therefore, we can find λ_i by solving $\mathrm{Det}(A - \lambda I) = 0$.

Eigenvectors and Inverse Matrices

- Eigenvector of a matrix $\boldsymbol{A}$ corresponding to eigenvalue λ_i given by

$$(\boldsymbol{A} - \lambda_i \boldsymbol{I})\, \boldsymbol{v}_i = \boldsymbol{0} \quad \text{or} \quad \boldsymbol{A}\boldsymbol{v}_i = \lambda_i \boldsymbol{v}_i$$

- Multiplying from the left with $\boldsymbol{A^{-1}}$ we obtain

$$\boldsymbol{v}_i = \lambda_i \boldsymbol{A}^{-1} \boldsymbol{v}_i$$

$$\leftrightarrow \boldsymbol{A}^{-1} \boldsymbol{v}_i = \frac{1}{\lambda_i} \boldsymbol{v}_i$$

- So, the eigenvector $\boldsymbol{v}_i$ of $\boldsymbol{A}$ is also an eigenvector of $\boldsymbol{A^{-1}}$ for the eigenvalue $1/\lambda_i$

Revisiting Eigenvectors and Eigenvalues

- Let‘s find the eigenvectors and eigenvalues of $A = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$ (Step 1).
- This means solving $\text{Det} \begin{bmatrix} 2-\lambda & 3 \\ 2 & 1-\lambda \end{bmatrix} = 0$.
- For a 2x2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the determinant is $ad - bc$. This gives us

$$\begin{aligned} (2-\lambda)(1-\lambda) - 6 &= \\ 2 - 2\lambda - \lambda + \lambda^2 - 6 &= \\ \lambda^2 - 3\lambda - 4 &= 0 \end{aligned}$$

- We solve with „pq formula“ $x^2 + px + q = 0 \Rightarrow x_{1,2} = -\frac{p}{2} \pm \sqrt{(\frac{p}{2})^2 - q}$:

$$\lambda_{1,2} = \frac{3}{2} \pm \sqrt{\frac{9}{4} + 4} = 1.5 \pm \sqrt{6,25}$$

$$\Rightarrow \lambda_1 = 4, \lambda_2 = -1$$

Revisiting Eigenvectors and Eigenvalues

- Let‘s find the eigenvectors and eigenvalues of $A = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$ (Step 2).
- With $\lambda_2 = -1$, we solve

$$\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix} = -1 \begin{bmatrix} v_{21} \\ v_{22} \end{bmatrix}$$

$$2v_{21} + 3v_{22} = -v_{21}$$

$$\Rightarrow 3v_{22} = -3v_{21}$$

$$\Rightarrow v_{21} = -v_{22}$$

- We know that eigenvectors only represent an orientation (the eigenvalue does the scaling), so we can freely choose a value for v_{22}. We will set $v_{22} = 1$, so $\vec{v}_2 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.
- Let‘s test: $\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -2+3 \\ -2+1 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = -1 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \lambda_2 \vec{v}_2$.

- **Let‘s find the eigenvectors and eigenvalues of** $A = \begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix}$ **(Step 2).**
- **With** $\lambda_1 = 4$ **, we solve**

$$\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix} = 4 \begin{bmatrix} v_{11} \\ v_{12} \end{bmatrix}$$

$$2v_{11} + 3v_{12} = 4v_{11}$$

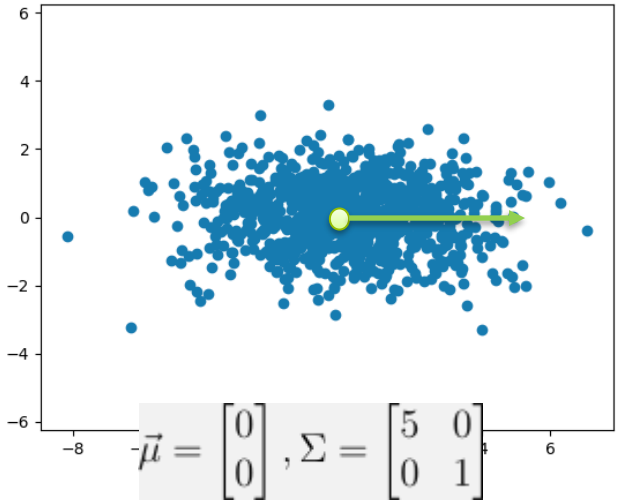
$$\Rightarrow 3v_{12} = 2v_{11}$$

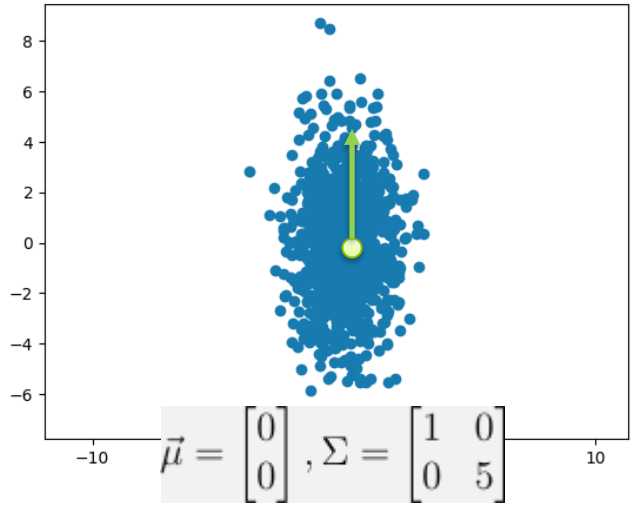
$$\Rightarrow v_{12} = \frac{2}{3} v_{11}$$

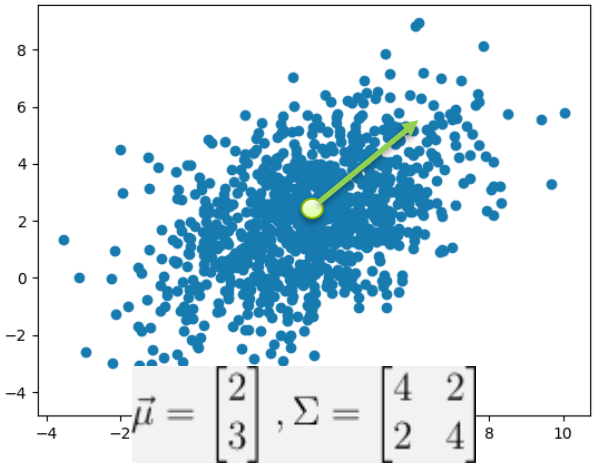
- **Again we can freely choose; we set** $v_{11} = 3$ **and get** $\vec{v}_1 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$ **.**
- **Check:**

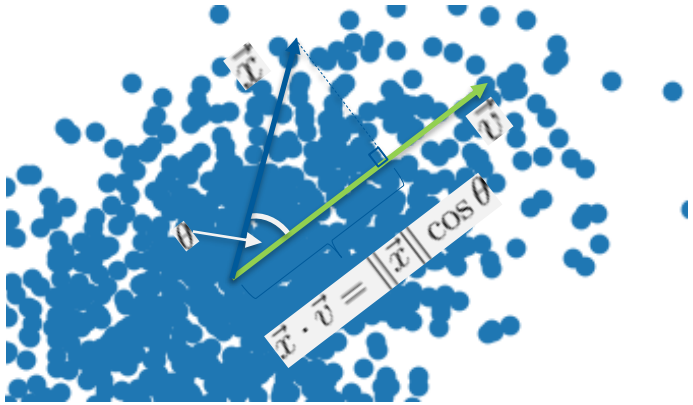
$$\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 6+6 \\ 6+2 \end{bmatrix} = \begin{bmatrix} 12 \\ 8 \end{bmatrix} = 4 \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Covariance and Eigenvalues









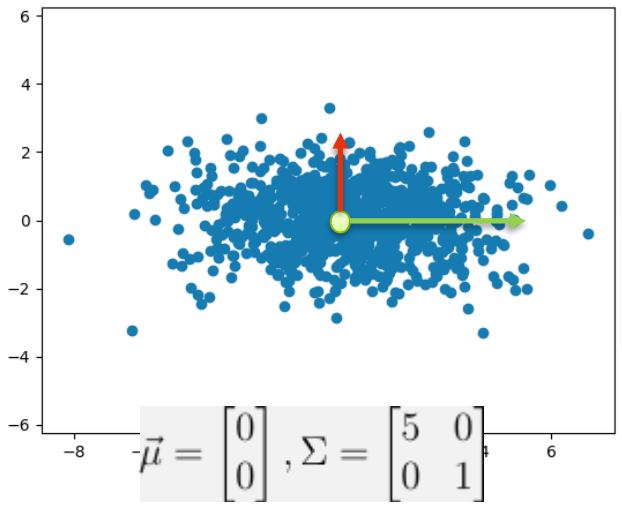
- The covariance matrix (Σ from now on) defines spread (variance) and orientation (covariance) of our data.
- If we want to represent the covariance matrix with a vector and its magnitude, we should try to find the vector that points into the direction of the largest spread of the data.
- The spread (variance) in this direction can give us the magnitude.
- Let‘s call that vector $\vec{v}$.
- We can project all of our data onto that vector with the dot product $\vec{v}^\top \vec{x} = \|\vec{x}\| \cos\theta$.
- The variance of the projected data is $\vec{v}^\top \Sigma \vec{v}$.
- We want to find the vector $\vec{v}$ that maximizes $\vec{v}^\top \Sigma \vec{v}$.

- **Maximizing any function of the form $\vec{v}^\top \Sigma \vec{v}$ with respect to $\vec{v}$ can be done with the so called *Rayleigh Quotient:***

$$R(M, \vec{x}) = \frac{\vec{x}^\top M \vec{x}}{\vec{x}^\top \vec{x}}$$

- **It can be shown that $R(M, \vec{x})$ reaches its maximum value $\lambda_{\max}$ when $\vec{x} = \vec{v}_{\max}$, the largest eigenvalue of M.**
- **Look up the proof if you're interested, but the important thing to remember is that this confirms our intuition:**
  - **The largest eigenvector of the covariance matrix always points in the direction of the largest variance,**
    and its magnitude equals the corresponding eigenvalue.
  - **The second largest eigenvector is always orthogonal to the largest eigenvector, and points into the direction of the second largest spread of data.**
  - ...and so on for all *n* dimensions of the matrix.

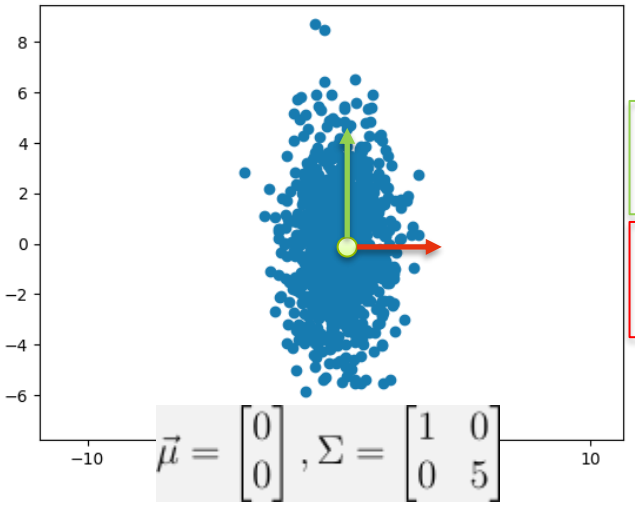
- **Let‘s look at some examples:**



$$\vec{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma = \begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\|v_1\| = \lambda_1 = 5, v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

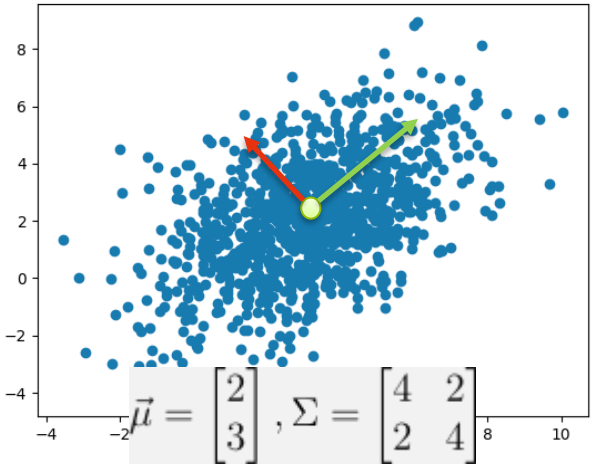
$$\|v_2\| = \lambda_2 = 1, v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



$$\vec{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix}$$

$$\|v_1\| = \lambda_1 = 5, v_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\|v_2\| = \lambda_2 = 1, v_2 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$



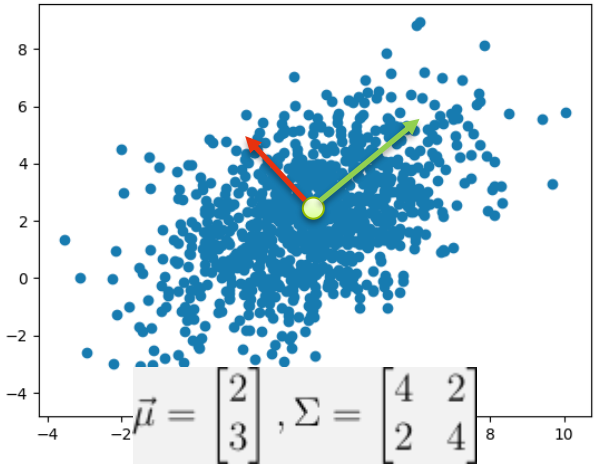
$$\vec{\mu} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$$

$$\|v_1\| = \lambda_1 = 6, v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\|v_2\| = \lambda_2 = 2, v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

It is intuitively clear that this one is obtained by a rotation of 45° around the imaginary z axis.

How can we derive this rotation mathematically, given a distribution of samples?

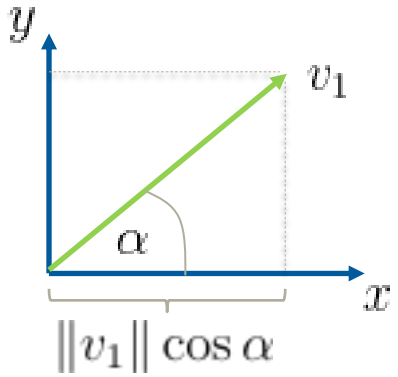


$$\vec{\mu} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$$

$$\|v_1\| = \lambda_1 = 6, v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\|v_2\| = \lambda_2 = 2, v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

The easy way is of course to look at the direction of the eigenvectors: –



$$\|v_1\| \cos \alpha = 1 \Rightarrow$$

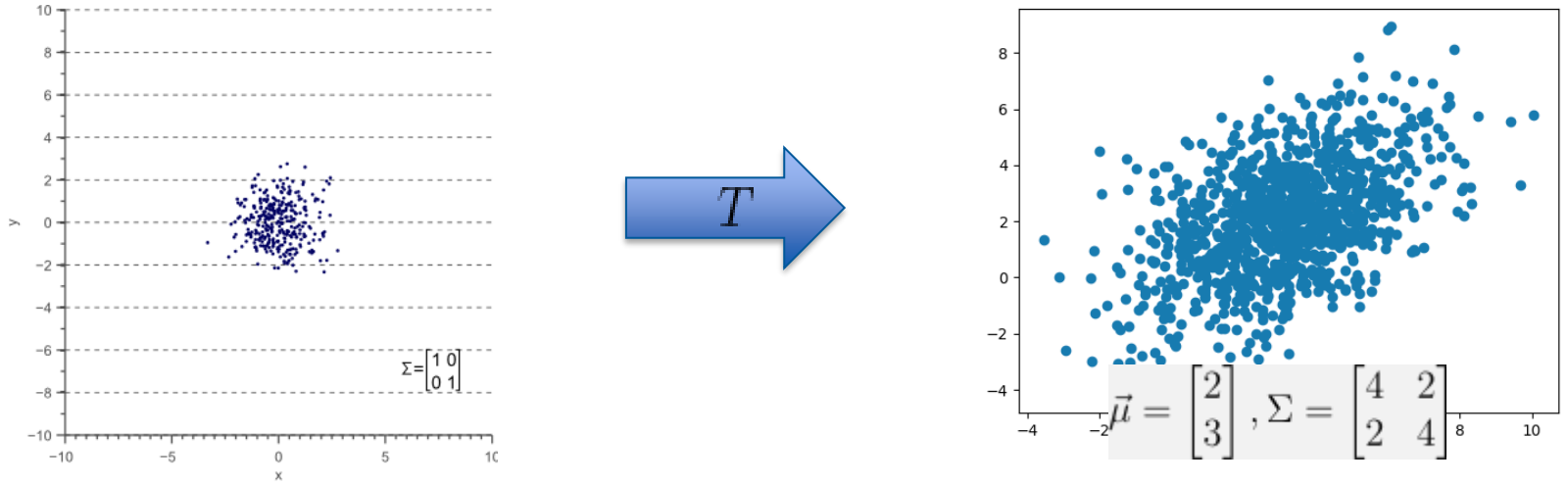
$$\alpha = cos^{-1} \frac{1}{\|v_1\|}$$

$$= cos^{-1} \frac{1}{\sqrt{2}}$$

$$= 45°$$

(or in code `atan2(v1y,v1x)`)

Covariance Matrix as a Linear Transformation



D $\quad D' = TD + \vec{\mu}$ $\quad D'$

where T is the product of a rotation matrix R and a scaling matrix S :

$$T = RS$$

Let‘s ignore the mean shift $\vec{\mu}$ for a while.

Unfortunately we can‘t solve for the Eigenvalues of D directly – they are complex, not real, numbers!

Why are Eigenvalues of a rotation matrix not real numbers?

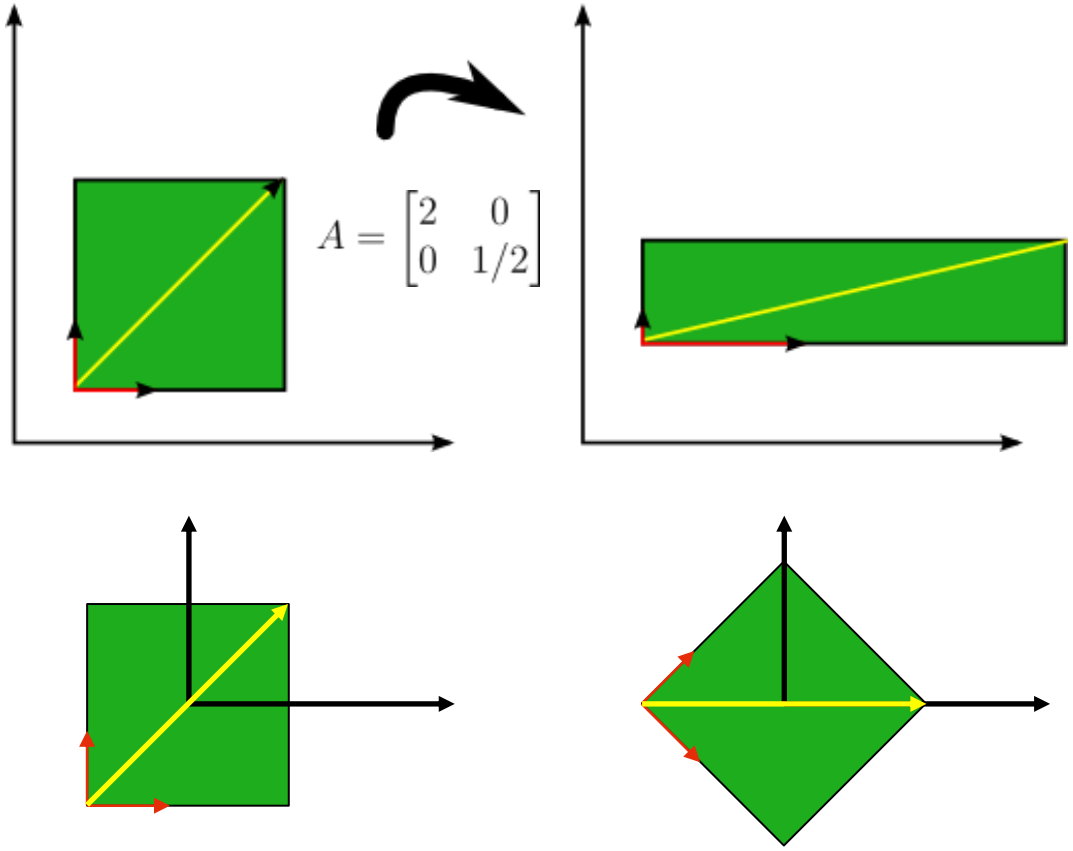
An eigenvector is a vector whose direction remains unchanged when a linear transformation is applied.

2D rotations change the directions of *all* vectors.

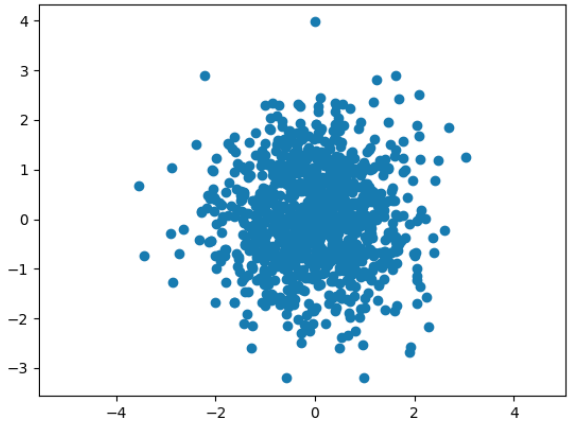
(3D rotations around individual axes do not – the rotation axis remains constant.)

Therefore while the determinant can be calculated, it does not yield real solutions.

We have to find another way.

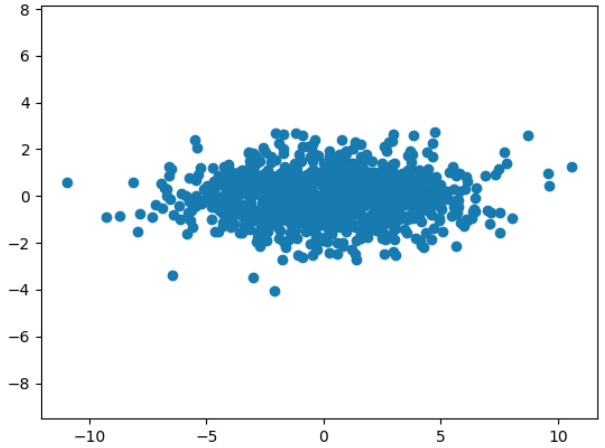


- In $D' = TD = RSD$, we know that $R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is a rotation matrix and $S = \begin{bmatrix} s_x & 0 \\ 0 & s_x \end{bmatrix}$ is a scaling matrix.
- Let‘s go step by step. From unity covariance,



$$\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

- we first scale the data by $\begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix}$.



$$\Sigma' = \begin{bmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_y^2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}$$

- We see that for scaling, the covariance Σ' of the scaled data D' is related to the scaling-only transform T in $D' = TD$ as $T = \sqrt{\Sigma'}$.

- What happens in the case of rotation?

- As we saw earlier, covariance can be expressed by eigenvectors and eigenvalues: $\Sigma\vec{v} = \lambda\vec{v}$
- For *n* dimensions we generally obtain up to *n* eigenvectors and eigenvalues. The resulting system can be written as

$\Sigma V = VL$, where $V = \begin{bmatrix} | & | & \dots & | \\ \vec{v}_1 & \vec{v}_2 & \dots & \vec{v}_n \\ | & | & \dots & | \end{bmatrix}$ is a matrix whose colums are the eigenvectors,

and $L = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$ is a diagonal matrix of eigenvalues.

- With $\Sigma V = VL$, and under the assumption that V is invertible, we can get

$$\Sigma = VLV^{-1}$$

  which is called the *eigendecomposition* of the covariance matrix.

- V and L can be computed using the Singular Value Decomposition algorithm [3].
- We already know that the column vectors of V represent the directions of variance in the data, and the eigenvalues in L represent the magnitude.
- This means V represents a rotation matrix, and $\sqrt{L}$ represents a scaling matrix.
- Renaming $V = R$ and $S = \sqrt{L}$ we can decompose further:

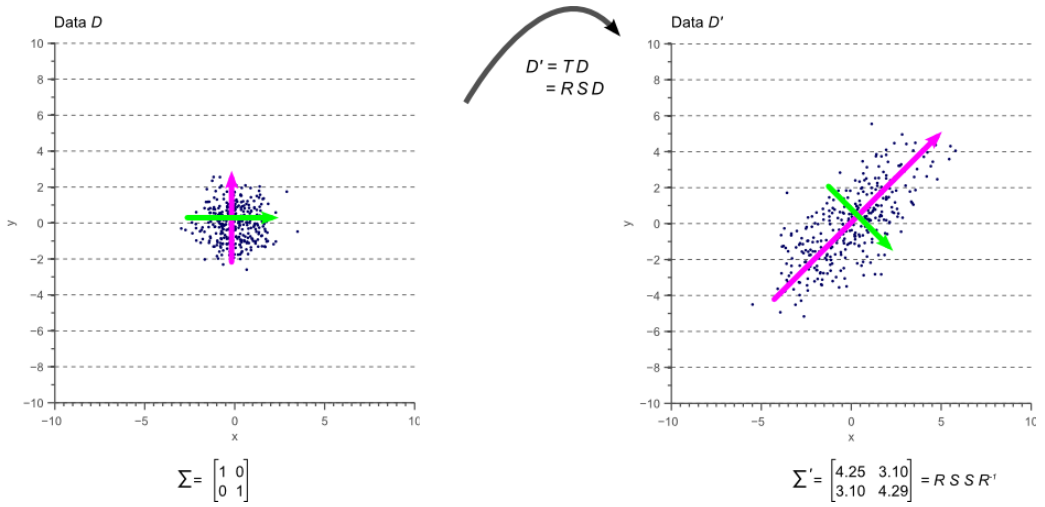
$$\Sigma = VLV^{-1} = RSSR^{-1}$$

- We also know that rotation matrices are orthogonal, i.e. $R^{-1} = R^{\top}$, and $S = S^{T}$ because the scaling matrix is a diagonal matrix.

$$\Sigma = RSSR^{-1} = RS(SR)^{\top}$$

- **We have defined the linear transformation transforming D into D' as $D' = TD = RSD$.**
- **In other words, if we apply the linear transformation defined by $T = RS$ to the original data D, we obtain the rotated and scaled data D' with covariance matrix**

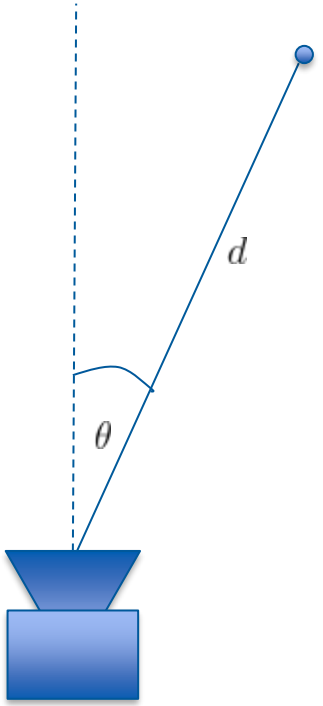
$$\Sigma' = RSSR^{-1} = RS(SR)^\top = TT^\top$$



Putting it all together – Sensor Modelling Example

- **Let‘s assume we have a sensor that delivers distance over azimuth.**
- **We take 1000 measurements at various angles and distances and find:**
  - all measurements witthin $\pm 3°$ across the whole field of view
  - all measurements within $\pm 1\%$ depending on distance.

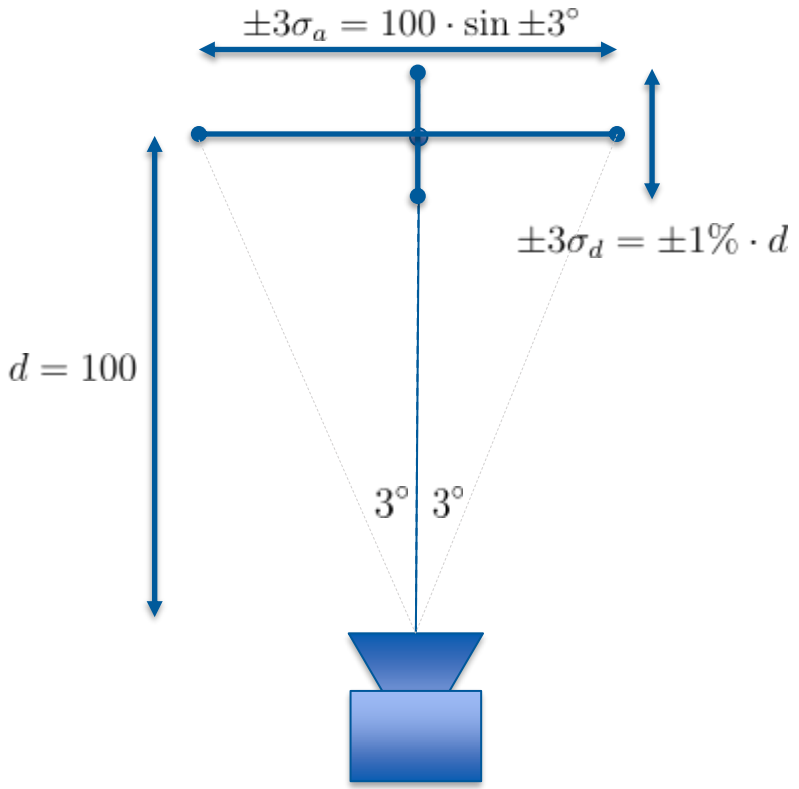
- **What is the covariance matrix of an object measured at 20° of azimuth, 100m of distance?**



- We start with an unrotated normal distribution of $\Sigma_0 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
- Since we empirically measured all measurements within $\pm 1\%$ of distance, we can assume $3\sigma_d = 100 \cdot 1\%$

  or $\sigma_d = \frac{1}{3}$ meters.
- Since we empirically measured all measurements with $\pm 3°$ of azimuth, we can assume $3\sigma_a = 100 \cdot \sin 3°$

  or $\sigma_a \approx \frac{5}{3}$ meters.
- So our unrotated data is scaled by $\begin{bmatrix} \frac{5}{3} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$.



- **We see the object at 20° of azimuth, so we need to rotate by**

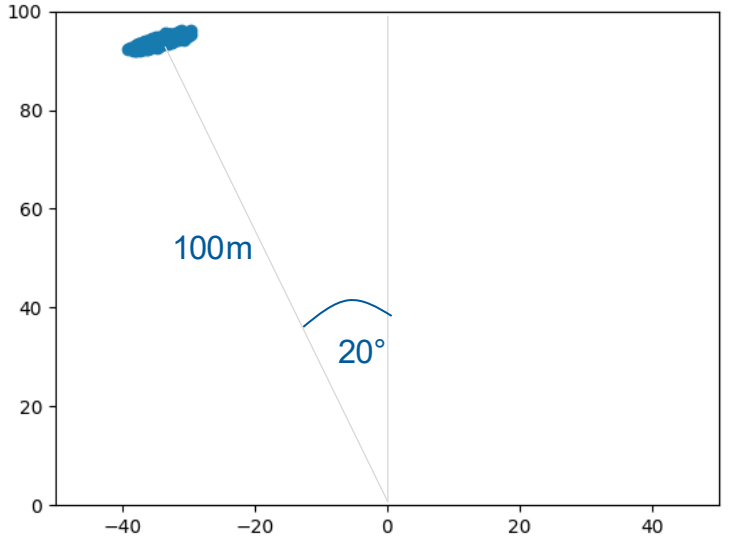
$$\begin{bmatrix} \cos 20° & -\sin 20° \\ \sin 20° & \cos 20° \end{bmatrix} \approx \begin{bmatrix} 0.94 & -0.342 \\ 0.342 & 0.94 \end{bmatrix}$$

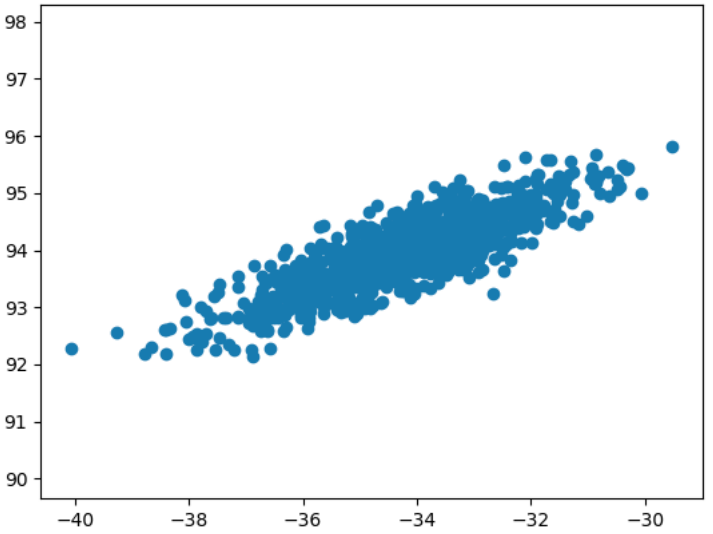
- **For a data transform of** $T = RS = \begin{bmatrix} \cos 20° & -\sin 20° \\ \sin 20° & \cos 20° \end{bmatrix} \begin{bmatrix} \frac{5}{3} & 0 \\ 0 & \frac{1}{3} \end{bmatrix}$

we obtain a covariance matrix of

$$\Sigma' = RS(SR)^T = \begin{bmatrix} \cos 20° & -\sin 20° \\ \sin 20° & \cos 20° \end{bmatrix} \begin{bmatrix} \frac{5}{3} & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} \frac{5}{3} & 0 \\ 0 & \frac{1}{3} \end{bmatrix} \begin{bmatrix} \cos 20° & \sin 20° \\ -\sin 20° & \cos 20° \end{bmatrix}$$

and can use this to generate a scatter plot of the multivariate normal distribution at $\vec{\mu} = \begin{bmatrix} 100 \sin \theta \\ 100 \cos \theta \end{bmatrix} \approx \begin{bmatrix} 34.2 \\ 93.97 \end{bmatrix}$ **.**





Multivariate Distributions

- Usually, a system is made of multiple states and multiple measurements
- Therefore, we consider vector-valued random variables $x \rightarrow \boldsymbol{x}$
- Then the variance becomes the covariance

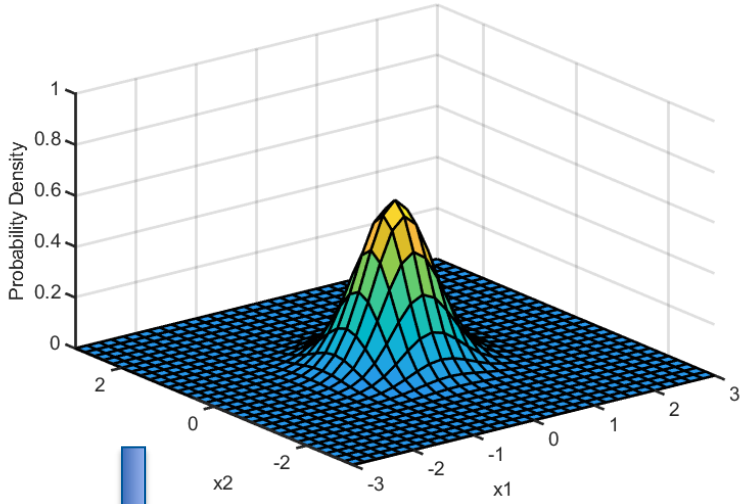
$$\mathrm{Var}[x] = E[(x - E[x])^2]$$

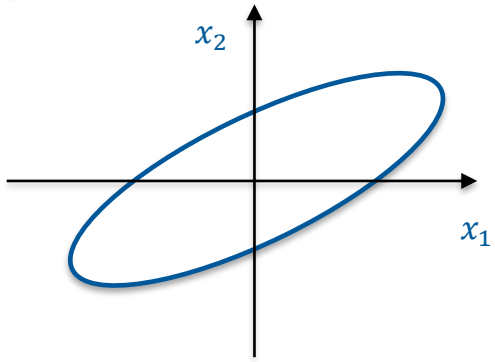
$$\Rightarrow \boldsymbol{P}_{XX} = \mathrm{Cov}\,(\boldsymbol{x}, \boldsymbol{x}) = \boldsymbol{E}\left[(\boldsymbol{x} - \boldsymbol{E}\,[\boldsymbol{x}]) \cdot (\boldsymbol{x} - \boldsymbol{E}\,[\boldsymbol{x}])^{\mathrm{T}}\right]$$

Multivariate Gaussian Distribution

- Notation $\boldsymbol{x} \sim N(\boldsymbol{\mu}, \boldsymbol{P}_{XX})$
- Probability density function

$$p_X(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n \det \boldsymbol{P}_{XX}}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\mathrm{T}} \boldsymbol{P}_{XX}^{-1} (\mathbf{x}-\boldsymbol{\mu})}$$





Two-dimensional Gaussian distribution

Gaussian Distributions: Summary

- Affine-Transformation: If $p(\boldsymbol{x})$ is Gaussian, $p(\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b})$ is also Gaussian
- Marginalization: If $p(\boldsymbol{x}, \boldsymbol{y})$ is Gaussian, $p(\boldsymbol{x})$ is also Gaussian
- Conditional distributions: If $p(\boldsymbol{x}, \boldsymbol{y})$ is Gaussian, $p(\boldsymbol{x}|\boldsymbol{y})$ is also Gaussian
- Product: If $p(\boldsymbol{x})$, $p(\boldsymbol{y})$ is Gaussian, $p(\boldsymbol{x}) \cdot p(\boldsymbol{y})$ is proportional to a Gauss-Distribution

These characteristics simplify computations involving Gaussian distributions. Most other distributions do not posses such advantageous characteristics.

Sources

1. **Mathematical derivation of Bessel‘s correction, from TileStats: https://www.youtube.com/watch?v=Bmwtyz7grqk**
2. **Vision Dummy: Eigenvalues and Eigenvectors, https://www.visiondummy.com/2014/03/eigenvalues-eigenvectors/**
3. **Singular Value Decomposition: https://en.wikipedia.org/wiki/Singular_value_decomposition**



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Thank you very much!